

随机模拟方法第四次作业

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1. 设 $X \sim \mathcal{U}[0, 1]$, 并且 f 单调. 求证: $\text{Cov}(f(X), f(1 - X)) \leq 0$.

解答: 由 $f(x)$ 的单调性可得

$$\begin{aligned}\text{Cov}(f(X), f(1 - X)) &= \mathbb{E}(f(X)f(1 - X)) - \mathbb{E}f(X)\mathbb{E}f(1 - X) \\&= \int_0^1 f(x)f(1 - x) \, dx - \int_0^1 f(x) \, dx \int_0^1 f(1 - x) \, dx \\&= \frac{1}{2} \int_0^1 \int_0^1 [f(x)f(1 - x) + f(y)f(1 - y) - f(x)f(1 - y) - f(y)f(1 - x)] \, dx \, dy \\&= \frac{1}{2} \int_0^1 \int_0^1 (f(x) - f(y))(f(1 - x) - f(1 - y)) \, dx \, dy \leq 0.\end{aligned}$$

□

2. 求证:

$$\text{Var}(f(X)) = \text{Var}(\mathbb{E}(f(X) \mid X^{(2)})) + \mathbb{E}(\text{Var}(f(X) \mid X^{(2)})).$$

解答: 下面证明更一般的命题: 设 X 和 Y 是任意两个随机变量 (向量), 则

$$\text{Var}(f(X)) = \text{Var}(\mathbb{E}(f(X) \mid Y)) + \mathbb{E}(\text{Var}(f(X) \mid Y)).$$

事实上,

$$\begin{aligned}&\text{Var}(\mathbb{E}(f(X) \mid Y)) + \mathbb{E}(\text{Var}(f(X) \mid Y)) \\&= \mathbb{E}(\mathbb{E}(f(X) \mid Y)^2) - \mathbb{E}(\mathbb{E}(f(X) \mid Y))^2 + \mathbb{E}(\mathbb{E}(f(X)^2 \mid Y) - \mathbb{E}(f(X) \mid Y)^2) \\&= \mathbb{E}(\mathbb{E}(f(X) \mid Y)^2) - \mathbb{E}(f(X))^2 + \mathbb{E}(f(X)^2) - \mathbb{E}(\mathbb{E}(f(X) \mid Y)^2) \\&= \mathbb{E}(f(X)^2) - \mathbb{E}(f(X))^2 \\&= \text{Var}(f(X)).\end{aligned}$$

□

3. 设 f, g 是概率密度函数且 $g(x) > 0$. 求证: 相对熵 $D(f||g) \geq 0$, 并且等号成立当且仅当 $f(x) = g(x)$.

解答: 由 Jensen 不等式,

$$\begin{aligned} D(f||g) &= \int_{\mathbb{R}} f(x) \ln \frac{f(x)}{g(x)} dx \\ &= - \int_{\mathbb{R}} f(x) \ln \frac{g(x)}{f(x)} dx \\ &\geq - \ln \left(\int_{\mathbb{R}} f(x) \cdot \frac{g(x)}{f(x)} dx \right) \\ &= - \ln \left(\int_{\mathbb{R}} g(x) dx \right) \\ &= 0, \end{aligned}$$

等号成立当且仅当 $\frac{f(x)}{g(x)} \equiv c$, 再结合 $\int_{\mathbb{R}} f(x) dx = \int_{\mathbb{R}} g(x) dx = 1$ 可得等价于 $f(x) = g(x)$. □